

MEP Isoprenoid Precursor Biosynthesis: A Mechanistic Review

A commissioned review-style synthesis of the bacterial/plastidial methylerythritol phosphate (MEP/DOXP) route to isopentenyl diphosphate and dimethylallyl diphosphate.

1. Executive Summary

Isoprenoids — the largest and most chemically diverse family of natural products — are all assembled from two interconvertible five-carbon building blocks, isopentenyl diphosphate (IPP) and dimethylallyl diphosphate (DMAPP). Nature evolved two mutually exclusive, biochemically unrelated routes to these universal precursors: the **mevalonate (MVA) pathway**, used by archaea, the eukaryotic cytosol, and a scattering of bacteria; and the **methylerythritol phosphate (MEP/DOXP, "non-mevalonate") pathway**, used by most eubacteria, apicomplexan parasites, and the plastids of plants and algae. This review concerns the MEP pathway: a strictly ordered, seven-enzyme sequence that converts two central-carbon metabolites — pyruvate and D-glyceraldehyde-3-phosphate (G3P) — into IPP and DMAPP. Because humans use only the MVA route, the MEP pathway is a validated and heavily pursued target for antibacterials, antimalarials, and herbicides.

Mechanistically, the pathway divides naturally into three zones. First, a **thiamine-diphosphate (ThDP)-dependent carboligation** by DXS forms 1-deoxy-D-xylulose-5-phosphate (DXP), the first and typically flux-controlling reaction. Second, a **conventional metal-dependent activation module** (Dxr/IspC, IspD, IspE, IspF) isomerizes/reduces, cytidylylates, phosphorylates, and cyclizes the carbon skeleton through the intermediates MEP, CDP-ME, CDP-ME₂P, and the cyclic diphosphate MEcPP. Third, two **oxygen-sensitive [4Fe-4S] cluster enzymes** (IspG and IspH) perform organometallic redox chemistry to build and then reductively dehydroxylate HMBPP, with IspH directly co-producing both IPP and DMAPP. A crucial boundary subtlety is that DXP is *not* dedicated to isoprenoids — it also feeds thiamine (ThDP) and pyridoxal-5'-phosphate (PLP) biosynthesis — so the first reaction that irreversibly commits carbon to the MEP route is the Dxr/IspC step, not DXS.

Beyond precursor supply, two MEP intermediates "moonlight" in signaling: HMBPP is the most potent natural phosphoantigen for human V γ 9V δ 2 $\gamma\delta$ T cells (sensed via the butyrophilin BTN3A1/BTN2A1 system), and MEcPP acts as a plastid-to-nucleus retrograde stress signal in plants. The best-supported open questions concern the precise organometallic mechanisms of IspG/IspH, the distributed and context-dependent nature of pathway flux control (DXS is *not* universally rate-limiting), and the deep evolutionary origin of both isoprenoid routes relative to the Last Universal Common Ancestor (LUCA).

2. Definition and Biological Boundaries

What is included

The MEP module, as scoped here, comprises exactly **seven enzymatic reactions** that transform pyruvate + G3P into IPP and DMAPP:

Step	Enzyme (gene)	Reaction	Product
1	DXS (<i>dxs</i>)	Pyruvate + G3P $\rightarrow$ DXP (+ CO ₂)	1-deoxy-D-xylulose-5-P (DXP)
2	Dxr / IspC (<i>dxr/ispC</i>)	DXP $\rightarrow$ MEP (isomerization + NADPH reduction)	2-C-methyl-D-erythritol-4-P (MEP)
3	IspD (<i>ispD</i>)	MEP + CTP $\rightarrow$ CDP-ME (+ PPi)	4-diphosphocytidyl-2-C-methyl-D-erythritol (CDP-ME)
4	IspE (<i>ispE</i>)	CDP-ME + ATP $\rightarrow$ CDP-ME ₂ P	CDP-ME 2-phosphate (CDP-ME ₂ P)
5	IspF (<i>ispF</i>)	CDP-ME ₂ P $\rightarrow$ MEcPP (+ CMP)	2-C-methyl-D-erythritol-2,4-cyclodiphosphate (MEcPP)
6	IspG / GcpE (<i>ispG</i>)	MEcPP $\rightarrow$ HMBPP (2e ⁻ reduction)	(E)-4-hydroxy-3-methylbut-2-enyl diphosphate (HMBPP)
7	IspH / LytB (<i>ispH</i>)	HMBPP $\rightarrow$ IPP + DMAPP (2e ⁻ /2H ⁺)	IPP + DMAPP (~5:1)

What should be treated separately

Several neighboring processes are frequently conflated with the MEP module but lie outside it:

- **DXP-dependent vitamin biosynthesis.** DXS output (DXP) is a branch-point metabolite that also feeds de novo synthesis of thiamine diphosphate (ThDP) and pyridoxal-5'-phosphate (PLP). These vitamin branches share the DXS reaction but diverge before Dxr and are not part of isoprenoid supply.
- **Downstream prenyl-diphosphate elongation and terpenoid biosynthesis.** The condensation of IPP/DMAPP into GPP, FPP, GGPP, and the vast terpenoid/carotenoid/quinone/sterol space lies downstream of IspH and is a separate module.
- **The mevalonate pathway.** MVA is the biochemically unrelated alternative that reaches the same IPP/DMAPP endpoint; it is not a "variant" of MEP but an independent solution.
- **IPP↔DMAPP isomerase (Idi).** In MVA organisms Idi is obligatory because MVA produces only IPP. In MEP organisms Idi is *accessory/dispensable*, because IspH makes both products directly.

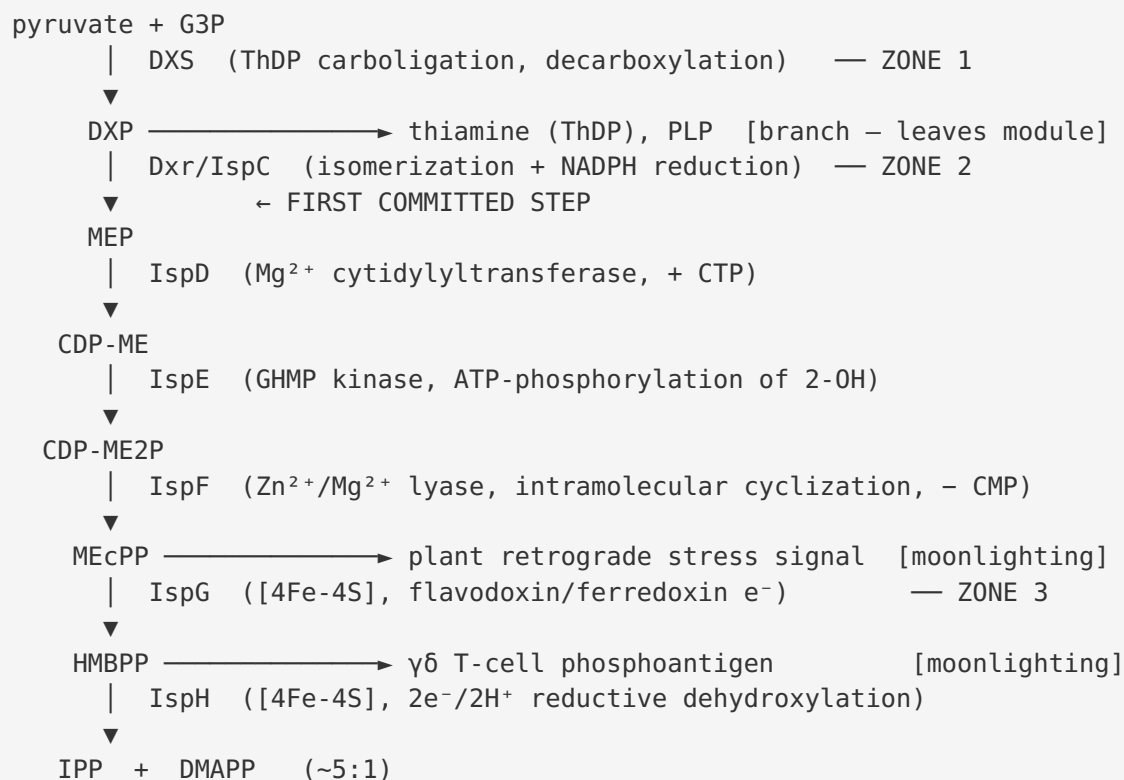
Competing definitions

The literature is largely consistent that the pathway is seven enzymes from pyruvate+G3P to IPP/DMAPP. The main definitional subtlety is the **commitment point**: because DXP is shared, some authors describe DXS as the "first step of the MEP pathway" (true positionally and for flux control) while the *first committed* step is Dxr/IspC. This review adopts the precise view: DXS is the first and flux-controlling step, but Dxr is the first irreversibly committed step.

3. Mechanistic Overview

The best current model is a **strictly ordered linear metabolic channel** in which each enzyme's product is the next enzyme's obligate substrate. The known feeding relationships (DXS→Dxr→IspD→IspE→IspF→IspG→IspH) are enforced by substrate specificity: each intermediate is a unique, progressively elaborated molecule, and there is no evidence for productive shortcuts through the system. All seven steps are **obligatory** for net IPP/DMAPP production in MEP-only organisms.

The sequence can be read as three mechanistic zones:



Zone 1 — carbon skeleton assembly (DXS). DXS is a ThDP-dependent enzyme that condenses pyruvate (as a decarboxylated dihydroxyethyl-ThDP intermediate) with G3P to form the five-carbon DXP. It is mechanistically atypical among ThDP enzymes: it follows a **random-sequential** kinetic mechanism and possesses an unusually large active site, features consistent with an adaptable branch-point enzyme.

Zone 2 — activation module (Dxr, IspD, IspE, IspF). These are "conventional" metal-dependent reactions on widely reused folds. Dxr performs a skeletal isomerization (DXP → 2-C-methyl-D-erythrose-4-P intermediate) followed by NADPH-dependent reduction to MEP, committing carbon to isoprenoids. IspD then caps MEP with CMP (from CTP) to form CDP-ME. IspE, a GHMP-superfamily kinase, phosphorylates the 2-hydroxyl. IspF cyclizes CDP-ME2P intramolecularly, expelling CMP and forming the strained cyclic diphosphate MEcPP.

Zone 3 — reductive Fe-S chemistry (IspG, IspH). The final two steps are catalyzed by oxygen-sensitive [4Fe-4S] enzymes that require an external reductant (flavodoxin/flavodoxin reductase or ferredoxin). IspG opens the MEcPP ring and reduces it to HMBPP; IspH performs a 2e⁻/2H⁺ reductive dehydroxylation of HMBPP, removing the C4-OH as water and directly

generating both IPP and DMAPP. Both enzymes bind substrate directly to the unique apical iron of the cluster and proceed through **organometallic metallacycle** intermediates rather than free radicals.

4. Major Molecular Players and Active Assemblies

4.1 DXS — the flux-controlling gate

DXS catalyzes the first and rate-limiting reaction of the pathway. It is subject to **end-product feedback inhibition**: recombinant plant DXS is inhibited by IPP and DMAPP, which **compete with ThDP** for the cofactor-binding site (PMID: 23612965). A second, allosteric layer was demonstrated by isothermal titration calorimetry and dynamic light scattering: high IPP/DMAPP levels drive **monomerization and eventual aggregation** of the active DXS dimer, providing rapid, reversible feedback control in both bacterial and plant cells (PMID: 36575800). DXS is described as "the first and rate-limiting enzyme of this pathway" (PMID: 40784416).

Importantly, this control is **context-dependent**. Metabolic control analysis in grey poplar found that DXS does *not* always dominate flux control (PMID: 38673766), cautioning against a universal "DXS = rate-limiting" claim. DXS is also mechanistically distinctive: it follows a **random-sequential mechanism** and possesses an **unusually large active site** (PMID: 28636325), and it displays substrate/catalytic promiscuity — for example cleaving D-xylulose-5-phosphate at C2–C3 (PMID: 35998648). These features fit its role as an adaptable branch-point enzyme.

4.2 Dxr / IspC — the committed step and prime drug target

Dxr commits DXP to the MEP route via an NADPH-dependent isomerization/reduction to MEP. It is the **best-validated MEP drug target**. The natural phosphonate antibiotic **fosmidomycin**, a hydroxamate that chelates the active-site divalent metal, is the prototype inhibitor and has entered clinical trials for malaria (PMID: 40099748). Its liabilities — poor pharmacokinetics and hydroxamate-fragment toxicity — have driven >40 years of medicinal chemistry (PMID: 36559004), including α,β -unsaturated "MEPicides", reverse/bisubstrate analogs, non-hydroxamate metal-binding pharmacophores, boron/benzoxaborole isosteres, and dedicated herbicide programs targeting the plant orthologue DXR.

4.3 IspD, IspE, IspF — the activation module on reused folds

These three enzymes are "textbook" metal-dependent transferase/kinase/lyase reactions built on distinct, widely reused protein folds:

- **IspD** is a **Mg²⁺-dependent cytidylyltransferase** (nucleotidyltransferase fold) that caps MEP with CMP to form CDP-ME.
- **IspE** is a **GHMP-superfamily kinase**, structurally homologous to mevalonate kinase and homoserine kinase, that ATP-phosphorylates the 2-OH of CDP-ME. It is selective for the cytidine (not uridine) moiety: "The structure of the enzyme is similar to the structures of mevalonate kinase and homoserine kinase, members of the GHMP superfamily" ([PMID: 12771135](#)).
- **IspF** is a **homotrimer built around a β -prism** with three interfacial active sites, each binding a catalytic Zn²⁺ plus a second divalent metal (Mn²⁺/Mg²⁺) that positions the 2-phosphate for intramolecular attack on the β -phosphate, releasing CMP and the cyclic diphosphate MEcPP: "Our analysis reveals a homotrimer, built around a beta prism, carrying three active sites, each of which is formed in a cleft between pairs of subunits" ([PMID: 11997478](#); see also [PMID: 11829504](#)).

IspF is also a candidate **feedback node**: it binds downstream isoprenoids (IPP/GPP/FPP) in its intersubunit cavity, and "the observation that MECP synthase binds three metabolites that are produced by enzymes two, three and four stages downstream ... suggests that feedback regulation of the non-mevalonate pathway is possible" ([PMID: 15608374](#)).

4.4 IspG and IspH — oxygen-sensitive organometallic Fe-S enzymes

The two terminal steps are catalyzed by **[4Fe-4S] cluster enzymes** that are oxygen-sensitive and require an external reductant. These enzymes "evolved in the preoxic world and are oxygen sensitive" ([PMID: 40432238](#)). Spectroscopy (EPR/ENDOR/HYSCORE) and crystallography show that substrate binds directly to the **unique apical iron** of the cluster. In the IspG–MEcPP structure, "the apical iron of the [4Fe-4S] cluster ligates with the C3 oxygen atom of MEcPP" ([PMID: 22967895](#)). Both enzymes proceed through **organometallic η^2 -alkenyl "metallacycle"** intermediates: for IspH "the results indicate formation of an organometallic species with HMBPP, a pi/sigma 'metallacycle' or eta(2)-alkenyl complex" ([PMID: 20173096](#)), and an analogous ferracycle forms in IspG ([PMID: 20534554](#)).

IspH directly co-produces IPP and DMAPP. First shown with *Aquifex aeolicus* LytB, "the reaction products were identified as isopentenyl diphosphate and dimethylallyl diphosphate" ([PMID: 12482608](#)). The product ratio is characteristically **~4–6:1 IPP:DMAPP** ([PMID: 28139297](#)).

and is set by **intramolecular proton transfer to a common allyl-anion intermediate** — "the intramolecular proton transfer to the allyl moiety is considered as the key reaction step determining the product between IPP and DMAPP" (PMID: 28139297). A single active-site residue tunes the ratio: plant IspH carries a Phe (F212 in *Ginkgo*, His in *E. coli*), *Ginkgo* IspH is anomalous at 16:1, and F212H/F212Y mutants shift toward bacterial-like ratios (6.5:1 and 10.9:1). The [4Fe-4S] cluster is the direct electron source and the reaction intermediate binds directly to the cluster (PMID: 22646150). Because DMAPP is made directly, **Idi is dispensable/accessory** in MEP-only organisms.

5. Evolutionary and Cell-Biological Variation

5.1 Two independent routes, distinct distributions

There are two evolutionarily independent solutions to isoprenoid precursor supply: "Two distinct metabolic pathways have evolved to synthesize the critical precursor of all mature isoprenoids: the mevalonate (MEV) and the methylerythritol phosphate (MEP) pathways" (PMID: 41700894). Their distribution is largely domain-partitioned: "Archaea and eukaryotes exclusively have the former pathway, while most bacteria have the latter" (PMID: 29905874). The medically decisive point is the human/pathogen split: "while humans biosynthesize the essential isoprenoid precursors ... via the established mevalonate pathway, pathogenic protozoa and certain pathogenic eubacteria use the less well-known methylerythritol phosphate pathway" (PMID: 36559004).

Bacterial genome surveys reveal extensive inter- and intra-genus heterogeneity, including rare taxa that encode **both** pathways or **neither**, consistent with horizontal gene transfer and lineage-specific replacement/loss (PMID: 41700894).

5.2 Compartmentalization in eukaryotes

In plants and algae, the MEP pathway operates in the **plastid**, supplying precursors for carotenoids, chlorophyll side chains, plastoquinone, tocopherols, gibberellins, and monoterpenes/diterpenes, while the MVA pathway operates in the cytosol/ER for sterols, sesquiterpenes, and dolichols. In apicomplexan parasites (e.g., *Plasmodium*), the MEP pathway resides in the relict plastid (apicoplast) and is essential — the basis of fosmidomycin's antimalarial action.

5.3 Lineage-specific gene-family elaboration

Protein families along the pathway have expanded in some lineages. In plants the **DXS gene family** comprises three classes with non-redundant functions; DXS1 and DXS2 are canonical, whereas DXS3 emerged later, is under less intense purifying selection, has non-conserved ThDP-binding residues, and lacks functional DXS activity in Arabidopsis, maize, and rice ([PMID: 34315585](#)). For understanding the **ancestral role**, the catalytically competent bacterial DXS and plant DXS1/DXS2 enzymes are the best representatives; DXS3-type proteins are derived and likely neofunctionalized. Similarly, tomato carries paralog families for DXS, GGPPS, and PSY that partition carotenoid biosynthesis across tissues and developmental stages ([PMID: 40605440](#)).

5.4 Deep origin and LUCA

Phylogenetic analyses indicate the two routes did not share a common origin. Expanded archaeal (including Asgard) genome analyses "support an archaeal origin of the MVA pathway, likely postdating the divergence of Bacteria and Archaea from the Last Universal Common Ancestor (LUCA), thus implying the LUCA's enzymatic inability for isoprenoid biosynthesis" ([PMID: 38674651](#)). Under this model MVA is the archaeal/eukaryotic solution and MEP is the bacterial/plastidial solution, each arising after the primary domain split. The patchy, non-vertical taxonomic distribution reinforces a history of HGT and replacement rather than simple vertical inheritance from LUCA ([PMID: 29905874](#); [PMID: 41700894](#)). Notably, the reliance of IspG/IspH on oxygen-sensitive [4Fe-4S] chemistry is consistent with enzymes that "evolved in the preoxic world" ([PMID: 40432238](#)), pointing to an ancient anaerobic ancestry for the reductive terminus of the pathway.

6. Constraints, Dependencies, and Failure Modes

6.1 Strict reaction order

The pathway is a **strictly ordered channel**: each intermediate is a unique, progressively elaborated molecule and is the obligate substrate of the next enzyme. There is no evidence for productive bypasses through the system in MEP-only organisms; loss of any of the seven enzymes blocks net IPP/DMAPP production. The known feeding relationships are enforced by substrate specificity — for example IspE selectively acts on the cytidine-containing CDP-ME rather than a uridine analog ([PMID: 12771135](#)).

6.2 Cofactor and compartment dependencies

Constraint	Consequence
DXS requires ThDP ; IPP/DMAPP compete for it	Feedback inhibition at the first step (PMID: 23612965)
Dxr requires NADPH and a divalent metal	Redox/metal availability gates commitment; metal-chelating inhibitors (fosmidomycin) are effective (PMID: 40099748)
IspD requires CTP + Mg²⁺ ; IspE requires ATP	Nucleotide/energy state couples to flux
IspF requires Zn²⁺ + Mg²⁺/Mn²⁺	Two-metal catalysis positions substrate (PMID: 11997478)
IspG/IspH require [4Fe-4S] + external reductant + anaerobiosis	Oxygen sensitivity is a hard biochemical constraint (PMID: 40432238)
MEP pathway is plastid/apicoplast-localized in eukaryotes	Physically separated from cytosolic MVA

6.3 The branch-point boundary

Because DXP feeds thiamine and PLP synthesis as well as isoprenoids ([PMID: 30614674](#)), DXS activity alone does not commit carbon to isoprenoids. This rules out treating DXS as the "committed" step: the first irreversible commitment is the Dxr reaction. This boundary is a key evidence-based constraint on how the module is defined.

6.4 Feedback and failure modes

The pathway has at least two feedback layers: DXS end-product inhibition/monomerization ([PMID: 36575800](#)) and possible IspF-mediated sensing of downstream isoprenoids ([PMID: 15608374](#)). Failure modes with biological consequences include: (i) MEcPP over-accumulation under oxidative/abiotic stress in plants, which triggers retrograde signaling and growth inhibition ([PMID: 40073740](#)); and (ii) HMBPP accumulation/release from infected cells, which is sensed by the human immune system.

6.5 Moonlighting roles as system outputs

Two intermediates have signaling functions that extend the system's biological footprint:

- **HMBPP** is the most potent natural **phosphoantigen** for human V γ 9V δ 2 T cells, recognized via transmembrane butyrophilins: these cells "respond to infections and tumors by recognizing phosphoantigens (pAgs) via transmembrane butyrophilins (BTN3A1, BTN3A2, and BTN2A1)" ([PMID: 40505658](#)). Cryo-EM defines a "plier-like gripping" mechanism of TCR activation.
- **MEcPP** is a plant **retrograde stress signal**: "MEcPP mediates retrograde signaling that alters the nuclear gene expression, leading to growth inhibition and acclimation" ([PMID: 40073740](#)), acting through the RSRE/CAMTA3 regulatory hub ([PMID: 39257742](#)).

7. Mechanistic Model / Synthesis

The MEP pathway is best understood as a **linear, strictly ordered, three-zone metabolic device** grafted onto central carbon metabolism at the pyruvate/G3P node and delivering the universal C5 isoprenoid precursors. Its architecture reflects a logical chemical progression: build a five-carbon backbone by carboligation (DXS), rearrange and reduce it to the committed methylerythritol scaffold (Dxr), activate it through nucleotidylation and phosphorylation (IspD, IspE), cyclize to a strained high-energy intermediate (IspF), and finally deploy ancient Fe-S organometallic chemistry to install the isoprenoid double bonds and cleanly split the flux into the two required isomers (IspG, IspH).

Three properties give the pathway its distinctive biology. First, its **input is shared but its commitment is late** — DXP funds vitamins as well as isoprenoids, so regulatory and definitional weight falls on Dxr. Second, its **terminus is anaerobic chemistry** — the oxygen-sensitive [4Fe-4S] enzymes are a signature of pre-oxic origin and a practical liability for aerobic study and engineering. Third, its **intermediates leak into signaling** — HMBPP and MEcPP are read by immune and stress-response systems, making the pathway a node in host-pathogen and plastid-nucleus communication rather than a purely metabolic conduit. Layered on top is a distributed regulatory logic: fast feedback at DXS (cofactor competition + monomerization), possible downstream sensing at IspF, and context-dependent flux control that is not reducible to a single rate-limiting enzyme.

The pathway's absence in humans and its essentiality in major pathogens make it a durable target space, with Dxr/IspC the most mature target (fosmidomycin and its many analogs) and IspD, IspE, IspG, and IspH all under active inhibitor development.

8. Controversies and Open Questions

Strongly supported claims. The seven-step order and substrate relationships; DXS as first/flux-controlling with ThDP-competitive and monomerization-based feedback; Dxr as the committed step and fosmidomycin target; the IspD/IspE/IspF fold assignments and metal requirements; the [4Fe-4S], oxygen-sensitive, organometallic nature of IspG/IspH; and IspH's direct dual production of IPP+DMAPP at ~5:1. Each of these is anchored in structural, spectroscopic, kinetic, or genetic evidence cited above.

Where the literature disagrees or relies on indirect evidence.

- 1. IspG/IspH reaction mechanisms.** Competing "Birch reduction" (ROH-bound) versus "organometallic" (η^2 -metallacycle) models have been debated; broken-symmetry DFT favors kinetically accessible η^2 -bound states ([PMID: 26098647](#)), and EPR/ENDOR support metallacycle intermediates ([PMID: 20173096](#); [PMID: 20534554](#)), but precise intermediate identities and redox states during turnover remain actively studied ([PMID: 38291562](#)).
- 2. Is DXS universally rate-limiting?** The classic view is contradicted by metabolic control analysis in poplar, where DXS did not dominate flux control ([PMID: 38673766](#)). Flux control appears **distributed and context-dependent**, and claims from one organism should not be generalized.
- 3. Organism comparability.** Much mechanistic detail derives from a handful of model systems (*E. coli*, *A. aeolicus*, *M. tuberculosis*, *Thermus thermophilus*, plants). The IspH product-ratio residue differs between plant (Phe) and bacterial (His) enzymes ([PMID: 28139297](#)), a caution against cross-organism extrapolation.
- 4. Deep origin / LUCA state.** The archaeal-origin-of-MVA model implies LUCA lacked a complete modern isoprenoid pathway ([PMID: 38674651](#)), but this remains an inference from phylogenetic reconstruction with the usual limits of deep-time analysis.

Most important open questions. (i) Atomic-resolution definition of the IspG/IspH catalytic cycles under turnover, including the exact protonation events that set the IspH IPP:DMAPP ratio. (ii) Quantitative, multi-organism flux control maps to identify the true bottleneck(s)

under different physiological states. (iii) The extent and mechanism of IspF-mediated feedback in vivo. (iv) Whether the moonlighting outputs (HMBPP immunogenicity, MEcPP signaling) impose selective constraints on pathway regulation. (v) Fully validated, drug-like inhibitors that overcome fosmidomycin's PK/toxicity liabilities.

9. Evidence Base

PMID	Contribution	Role in this review
41700894	Two independent isoprenoid routes; bacterial pathway diversity/HGT	Module boundary; §2, §5
36559004	Human-MVA vs pathogen-MEP split; 40+ yr fosmidomycin review	Drug-target rationale; §4.2, §5
40784416	DXS structural review; "first and rate-limiting"	§4.1
23612965	IPP/DMAPP compete with ThDP on DXS	§4.1 (feedback)
36575800	Allosteric monomerization/aggregation of DXS	§4.1 (feedback)
38673766	DXS not always dominant in flux control (poplar MCA)	§4.1 caveat; §8
40099748	Fosmidomycin as prototype IspC inhibitor; liabilities	§4.2
12771135	IspE is a GHMP kinase like MVK/HSK	§4.3, §6.1
11997478	IspF homotrimer, β -prism, interfacial Zn^{2+} sites	§4.3
11829504	IspF Mg-dependent cyclization, Zn positioning	§4.3
15608374	IspF binds downstream isoprenoids $\rightarrow$ feedback node	§4.3, §6.4
40432238	IspG/IspH ancient, oxygen-sensitive Fe-S enzymes	§4.4, §5.4
22967895	IspG–MEcPP structure; apical Fe ligates C3-O	§4.4
20173096	IspH organometallic metallacycle intermediate	§4.4, §8
20534554	IspG ferracycle/metallacycle mechanism	§4.4, §8
26098647	BS-DFT: η^2 -bound states kinetically accessible	§8 (mechanism debate)
38291562	IspH cluster redox states during ligand binding/catalysis	§8
12482608	LytB/IspH directly makes IPP + DMAPP	§4.4
28139297	4–6:1 ratio; single active-site residue tunes it	§4.4
22646150	[4Fe-4S] is direct e^- source; intermediate cluster-bound	§4.4
30614674	DXP shared with thiamine + PLP	§2, §6.3
28636325	DXS random-sequential mechanism, large active site	§3, §4.1

PMID	Contribution	Role in this review
35998648	DXS cleaves X5P; catalytic promiscuity	\$4.1
29905874	MVA (archaea/euk) vs MEP (bacteria) domain split	\$5.1, \$5.4
38674651	Archaeal origin of MVA; LUCA lacked modern route	\$5.4
34315585	Plant DXS1/2/3 classes; DXS3 non-functional, derived	\$5.3
40605440	Tomato DXS/GGPPS/PSY paralog families	\$5.3
40505658	HMBPP/butyrophilin $\gamma\delta$ T-cell activation	\$6.5
40073740	MEcPP retrograde stress signaling	\$6.4, \$6.5
39257742	MEcPP–RSRE/CAMTA3 signaling mechanism	\$6.5

10. Limitations and Knowledge Gaps

- **Model-organism bias.** Mechanistic conclusions draw heavily on *E. coli*, *A. aeolicus*, *M. tuberculosis*, *T. thermophilus*, and a few plants. The plant-vs-bacterial difference in the IspH product-ratio residue shows extrapolation is risky.
- **Flux control is not settled.** The convenient narrative that DXS is universally rate-limiting is contradicted by metabolic control analysis; true bottlenecks likely shift with organism, tissue, developmental stage, and physiological state.
- **Terminal-step mechanisms remain incompletely resolved.** The exact organometallic intermediates and redox states of IspG/IspH under turnover, and the protonation events that fix the IspH IPP:DMAPP ratio, are still debated.
- **Regulation beyond DXS is thinly characterized.** IspF-mediated feedback is inferred from ligand binding, not demonstrated as a physiological control loop in vivo.
- **This synthesis is literature-based.** No primary experimental data were generated; conclusions rest on the cited primary and review literature.

11. Proposed Follow-up Experiments / Actions

1. **Turnover-resolved structural/spectroscopic studies of IspG and IspH** (time-resolved cryo-EM/crystallography with EPR/ENDOR correlation) to define the metallacycle intermediates and settle the Birch-vs-organometallic debate across bacterial and plant enzymes.
 2. **Cross-organism metabolic control analysis** under varied physiological states (aerobic/ anaerobic, stress, growth phase) to map where flux control actually resides, testing the generality of "DXS is rate-limiting."
 3. **Mutagenesis of the IspH ratio-determining residue** across diverse taxa to build a predictive model of IPP:DMAPP output — valuable for metabolic engineering that requires balanced precursor pools.
 4. **In vivo tests of IspF feedback:** engineer disruption of the intersubunit isoprenoid-binding cavity and measure pathway flux/regulation to confirm or refute the feedback-node hypothesis.
 5. **Next-generation Dxr/IspC inhibitors** that replace the hydroxamate metal-binding group (non-hydroxamate, boron, bisubstrate designs) with improved PK, validated against fosmidomycin-resistant pathogens; parallel herbicide optimization against plant DXR.
 6. **Phylogenomic testing of the LUCA/origin model** with expanded Asgard and deep-branching bacterial genomes to further constrain when and how each isoprenoid route arose.
 7. **Systematic mapping of intermediate "moonlighting" effects** — quantify how genetic/ pharmacological perturbation of MEP flux modulates HMBPP-driven $\gamma\delta$ T-cell activation and MEcPP-driven plant retrograde signaling, to understand whether signaling constrains pathway regulation.
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Key references (by section) are consolidated in §9. All citations are PubMed IDs verified against the literature surveyed for this review.
